EPISODE 8

Aptitude & Reasoning

Quantitative, Logical & Verbal for Interviews

Percentage & Profit	Time, Work & Speed
Probability & Combinations	Progressions
Logical Reasoning	Verbal Reasoning

APTITUDE QUICK REFERENCE — FORMULAS & TRICKS

Aptitude Topics — Quick Reference

- **Number Systems** — Divisibility, HCF, LCM, Factors
- **Percentage & Profit** — Cost/Selling price, Markup, Discount
- **Time & Work** — Efficiency, Pipes & Cisterns
- **Speed & Distance** — Relative speed, Trains Boats
- **Interest** — SI & CI formulas Compounding
- **Probability** — P(E)=f/n Combinations Permutations
- **Ratio & Proportion** — Direct, Inverse, Mixtures
- **Progressions** — AP & GP Formulas, Series sum
- **Calendar & Clocks** — Odd days, Hand angles, Day of week
- **Averages** — Weighted, Group data, Missing no.

Master these formulas and trick patterns to crack any aptitude round

Essential Formulas

Topic	Formula	Example
Simple Interest	SI = (P × R × T) / 100	P=1000,R=5%,T=2 → SI=100
Compound Interest	CI = P(1+R/100)■ – P	P=1000,R=10%,T=2 → CI=210
Speed-Dist-Time	S = D/T, D = S×T, T = D/S	60km/h × 2h = 120km
Work	1/A + 1/B = 1/T (combined)	A=6d, B=12d → T=4d
Percentage change	((New–Old)/Old) × 100	50→60 = +20%
Profit %	((SP–CP)/CP) × 100	CP=80, SP=100 → 25%
HCF×LCM	HCF × LCM = Product of two numbers	HCF(12,18)=6, LCM=36, 6×36=216
Permutation	P(n,r) = n!/(n-r)!	P(5,2)=20
Combination	C(n,r) = n!/(r!(n-r)!)	C(5,2)=10
Probability	P(E) = Favourable / Total	P(Head)=1/2

LOGICAL REASONING — PATTERNS & TRICKS

Type	Strategy	Watch For
Direction Sense	Draw compass, track each turn	Final displacement ≠ total distance
Blood Relations	Draw family tree diagram	Paternal vs maternal side
Syllogism	Venn diagram method	'Some' = at least one, 'All' = every
Series	Find difference pattern (1st, 2nd order)	Alternating series, squares, cubes
Arrangements	Fix reference point, apply constraints	'Together' = treat as single unit
Clocks	Relative speed = 5.5° per min	At t mins: angle = |30H - 5.5M|
Calendar	Count odd days (7 days = 0 odd days)	Leap year = 2 odd days, else 1

Speed Tricks for Aptitude Exams

Percentage: x% of y = y% of x (e.g., 8% of 25 = 25% of 8 = 2)
Multiplication: 48×52 = (50-2)(50+2) = 2500-4 = 2496
Train crossing pole: Time = Length / Speed
Train crossing platform: Time = (Length + Platform) / Speed
Boat upstream: Speed = Boat - Stream; Downstream = Boat + Stream
Work shortcut: If A does in 'a' days, rate = 1/a per day
Compound Interest shortcut: For 2 years: CI = SI + SI^2/100P
Number of factors: prime factorize then (e1+1)(e2+1)... formula

VERBAL REASONING — SENTENCE SKILLS

Skill	Strategy	Common Errors
Sentence Ordering	Find the topic sentence (usually S1), then logical flow	Confusing cause and effect order
Error Spotting	Check: Subject-verb agreement, Tense consistency	Much/Many', 'fewer/less' confusion
Sentence Improvement	Prefer active voice, concise phrasing	Redundancy: 'return back', 'future plans'
Reading Comprehension	Read question first, then scan passage	Don't infer beyond what's stated

Episode 8 Summary

✓ Classic Chapters: Percentage, Profit/Loss, SI, CI, Ratio, Average
✓ Speed/Work: Time-Distance, Trains, Boats, Pipes & Cisterns
✓ Probability & Combinations: P(n,r), C(n,r), basic probability
✓ Progressions: AP (nth term = a+(n-1)d), GP (nth term = ar■■^{1})
✓ Logical Reasoning: Direction, Blood Relations, Syllogism, Series
✓ Calendar & Clocks: Odd days method, clock angle formula
✓ Verbal: Sentence ordering, error spotting, improvement techniques